

Technical Whitepaper: Uli SDK

Bridging Heterogeneous Assets with Agentic AI via Dynamic Knowledge Graphs and Enhanced Cursor-on-Target (CoT)

Executive Overview

The **Uli (Unified Link Interface) SDK** serves as the tactical software nervous system for Agentic AI, operating as a high-performance middleware that connects heterogeneous physical assets—including ground platforms, aerial units, and edge sensors—with multi-domain AI reasoning ecosystems. Built upon a modular **client-server messaging architecture** where distributed services interact over a standardized Unified Link Interface, Uli SDK enables dynamic discovery, asset contextualization, and runtime adaptability across contested environments. By synthesizing asset identity, functional capabilities, and live telemetry feeds into a structured **Knowledge Graph Context Layer**, Uli SDK delivers a shared, real-time world model for AI agents and human operators. Enhanced by decentralized, service-hosted **UDP Multicast Cursor-on-Target (CoT)** event pipelines, Uli SDK transports multi-modal situational awareness—including spatial XML, low-latency imagery, 3D point cloud meshes, and neural tensor arrays—with zero cloud dependency.

1. Core Architectural Principles

Dynamic Client-Server Messaging Architecture

At the core of the Uli SDK is a distributed **client-server messaging architecture** where software components expose and consume services through a standardized Unified Link Interface:

- **Dynamic Service Lifecycle:** Service-hosting components can be registered, initialized, reconfigured, or decommissioned at runtime without disrupting ongoing operations or requiring system-wide recompilation.
- **Runtime Adaptability:** The architecture dynamically adjusts to changing mission requirements, payload reconfigurations, and intermittent tactical network connectivity.
- **Modular Open Systems Approach (MOSA):** Engineered in full compliance with MOSA principles, eliminating vendor lock-in and allowing modular integration of specialized AI modules, robotics controllers, and tactical sensors.
- **Reduced Integration Latency:** Eliminates rigid, hard-coded payload integrations, cutting onboarding and reconfiguration timelines by **40%+**.

Multi-Domain Heterogeneous Interoperability

Engineered in compliance with the **DoD Modular Open Systems Approach (MOSA)**, the Uli SDK provides standard data fabrics and APIs across disparate operating domains:

- **Unmanned Ground & Aerial Systems:** Autonomous mobility, sensor suites, and payload controllers.
- **Industrial & Tactical Systems:** Heavy vehicle buses (J1939), robotics stacks (ROS/ROS2, NVIDIA ISAAC ROS), and edge compute clusters.
- **Tactical Command & Intelligence Layers:** Next-Generation Command and Control (NGC2) frameworks, JADC2 nodes, TAK/CoT networks, and Agentic AI reasoning engines.

2. Knowledge Graph Synthesis & AI Reasoning

The Uli SDK implements a **Discovery-Driven Semantic Engine** that transforms unstructured telemetry and hardware specifications into contextual knowledge optimized for Large Language Models (LLMs) and Agentic reasoning engines.

Client-Server Context Discovery Vectors

Through dedicated discovery services hosted across Uli SDK endpoints, assets self-announce three foundational context vectors:

- **Asset Context:** Unique system identification (URI, serial, hardware profile), spatial localization/pose, operational domain, and security privileges.
- **Capability Context:** Executable functional skill sets, operational boundaries, command interfaces, and acceptable parameter envelopes.
- **Telemetry Context:** Data topic definitions, streaming rates, engineering units, and schema definitions.

Dynamic Knowledge Graph Synthesis

Context vectors discovered by client-server endpoints are ingested as nodes and relationships within an edge-hosted **Knowledge Graph**. This generates an operational world model that allows AI agents to query asset availability, perform semantic state estimation, and evaluate mission plans against live environmental states in real time.

Autonomous Command & Execution

AI agents leverage the Knowledge Graph to formulate precise task allocations, mission plans, and low-level control parameters. The Uli SDK translates high-level agentic intent into

high-fidelity commands, driving physical actuation across ROS2 nodes, bare-metal C++ loops, and payload controllers.

3. Extended Cursor-on-Target (CoT) & Human-Machine Teaming

Uli SDK implements tactical situation sharing directly within its client-server messaging architecture, enabling any component to host a **CoT Event Service** for distributed broadcasting.

Flexible CoT Service Hosting: Any node or subsystem running Uli SDK can instantiate a lightweight CoT Event Service to originate, transform, and broadcast situational updates without requiring a central broker.

UDP Multicast Tactical Transport: Utilizes bandwidth-efficient UDP Multicast streams to broadcast CoT events across tactical edge networks, ensuring simultaneous delivery to AI agents, edge nodes, and operator end-user devices (EUDs).

Multi-Modal Payload Containers: Extends traditional spatial CoT XML metadata to encapsulate high-bandwidth binary buffers directly within event packets:

- Low-latency EO/IR imagery frames.
- 3D spatial point clouds and mesh geometry.
- Neural network tensor arrays and latent feature embeddings.

Collaborative AI Tensor Sharing: Distributed AI nodes exchange neural feature maps over the CoT transport layer, enabling swarm-level collaborative inference and state prediction without streaming saturated raw video feeds.

TAK Ecosystem Interoperability: Provides bi-directional compatibility with ATAK, WinTAK, and TAK Server architectures, delivering a shared Common Operational Picture (COP) for human operators and AI agents.

Agent-to-UI (A2UI) Dynamic Interfaces: Native Dart-FFI and C++ bindings empower AI agents to dynamically generate and update Flutter control dashboards as newly discovered asset services come online.

4. Operational Integrity: Security, Safety, and Protocol Support

Security Architecture

- **Certificate-Based Authentication:** Cryptographic identity verification for every node joining the Uli SDK context mesh.

- **Role-Based Access Control (RBAC):** Strict enforcement of Data Access Privileges (Classified, Controlled, Unclassified) and Application Roles (Operator, Maintainer, Administrator).

Safety & State Reliability

- **Redundant Safety Triggers:** Automated monitoring coupled with hardware-level emergency stop (E-Stop) commands.
- **Two-Tiered State Machines:** Deterministic operational modes (Standard, Reduced, Training, Maintenance) to prevent unvalidated command execution during critical failures.

5. Development and Deployment Ecosystem

- **Multi-Language Bindings:** Automated code generation tools produce native C++ implementations, high-efficiency Python bindings for ML pipelines, and Dart-FFI bindings for cross-platform UI rendering.
- **Cross-Compilation:** Built using Bazel for optimized target compilation across x86_64 servers and ARM64 NVIDIA Jetson platforms (Nano, Xavier, Orin, DGX Spark).
- **Reference Platform (Uli Kaya):** Validated on Open Vision Technology's **Uli Kaya** omnidirectional 3-wheel development robot, demonstrating full-stack sensor fusion, real-time CoT broadcasting, and dynamic agentic control execution.

Conclusion

The Uli SDK delivers an adaptable software architecture for next-generation defense and autonomous enterprise systems. By combining dynamic Knowledge Graph context generation, high-bandwidth UDP Multicast CoT extensions, and MOSA-compliant protocols, Uli SDK bridges physical sensors and edge hardware with Agentic AI—delivering actionable situational awareness and autonomous execution across multi-domain operational environments.

For technical inquiries and developer access, visit ulisdk.com.